VA DIVISION A • 3RD GRADE

Tower Building

Official Study Guide • 2025–2026 Season

- Official Materials
- Scoring Formula
- Triangulation
- Straw Column Design
- Stability
- Written Assessment

Event Overview — Official Rules

Rules

Build the tallest freestanding tower possible using ONLY paper straws and masking tape, capable of holding a 300-gram mass for 20 seconds. You will also take a written assessment.

WHAT YOU GET

- 30 paper drinking straws (~20 cm each)
- 1 meter of 2.54 cm masking tape
- A 300-gram mass for testing
- A 5×5×0.6 cm plywood block (placed on top)
- Scissors (provided at event)
- Event supervisor records unused straws

BUILD RULES

- 45-minute total session (written test + build)
- Tower must hold the 300g mass for 20 seconds
- Tower must have a flat area at top for the 5×5 cm block
- No sticky tape may touch the flat surface
- All support parts must rest on the flat surface
- Straws may be cut or left whole — your choice

SAFETY REQUIRED

- ANSI Z87+ impact-rated safety glasses required
- Must be worn during any competitor's testing phase
- Eyeglasses are NOT safety rated
- Without proper eye protection = no-show score
- Teams must bring their own safety glasses
- Writing utensils required (bring pencil/pen)

🏅 Scoring Formula — How Points Are Calculated

Scoring

Final Score (FS) = Assessment Score + Height Score. Maximum total = 100 points. Both components are worth 50 points each.

Term	Formula	Notes
Final Score	FS = Assessment Score (TAS) + Height Score (THS)	Max 100 points total
Assessment	TAS = (your score ÷ best score among all teams) × 50	Max 50 pts
Height Score	THS = (your height ÷ tallest height among all teams) × 50	Max 50 pts
Tiebreaker 1	Time to structural failure (longer = better)	After primary score
Tiebreaker 2	Fewest unused straws — use your materials!	Second tiebreaker
Tiebreaker 3	Best assessment score	Final tiebreaker

Example: Best height = 450mm. Your height = 375mm → Height score = (375÷450)×50 = 41.67 pts. Best assessment = 9, you score 4 → Assessment score = (4÷9)×50 = 22.22 pts. Final = 63.89

Tower Testing Phase — Exact Procedur

Testing

1 Set Tower on Flat Surface

All support parts of the tower must rest on the flat even surface. No sticky tape may touch the flat surface.

2 Event Supervisor Measures Height

Height is measured to the nearest millimeter BEFORE the load is placed. The height of the wood block is NOT added to the tower's measurement.

3 Place the Wood Block

The team places the 5×5×0.6 cm plywood block on top of their structure. The block must fit on the tower's top platform.

4 Place the 300g Mass

The team places the 300-gram mass on top of the wood block and removes their hand. Timing begins the moment their hand leaves the mass.

5 20-Second Hold

Testing ends when: (a) the tower fails/collapses, or (b) 20 seconds elapse. Event Supervisor is the official timekeeper, recorded to the nearest 1/100th of a second.

6 Event Supervisor Records

Records: Height (mm), time to failure (1/100 sec), number of unused straws, team assessment score.

Triangulation — The Core Principle

Triangles

A triangle cannot change shape without changing the length of its sides. This makes it the ONLY rigid polygon — the foundation of every strong structure built from straws.

WHY SQUARES FAIL

- A square frame has 4 joints — each one can rotate
- Under sideways (shear) force, a square collapses into a rhombus
- Without a diagonal brace, your straw frame will fold immediately
- Even tight tape joints will eventually rotate under load
- RULE: Never leave a rectangular frame unbraced

HOW TRIANGULATION WORKS

- Add one diagonal straw across any rectangular frame → creates 2 triangles
- Each triangle member carries only tension OR compression — not bending
- The force in each member is distributed along its full length
- A triangulated frame is rigid in all directions within its plane
- Add diagonals on ALL faces of a 3D tower for full rigidity

STRAW TRIANGULATION TIPS

- Diagonal braces: cut straws at 45° and tape at both ends
- Use short diagonals at each tier level — not one long diagonal across whole tower
- The more triangles, the stiffer — but each uses material
- Prioritize the LOWER tiers: base failures collapse everything above
- Cross-bracing (X pattern) is strongest but uses 2 straws

🥤 Straw Columns — Buckling & Bundling

Columns

Paper straws are hollow cylinders — strong in compression along their length, but vulnerable to BUCKLING (bending sideways). Your main job is preventing buckling while maximizing height.

BUCKLING PREVENTION

- Short columns buckle less — add lateral bracing at each tier
- Euler's Law: halving column length = 4× more buckling resistance
- A 20cm straw buckles easily — brace it at 10cm intervals
- Taping two straws together (bundle) roughly doubles strength
- Three straws bundled in a triangle cross-section = much stiffer

JOINT TECHNIQUES

- Overlap straws 2–3 cm before taping — more surface = stronger
- Wrap tape AROUND the joint for compression band effect
- Tape both sides of every joint — front and back
- T-joints (straw into straw side) are weakest — reinforce with extra wrap
- Use small pieces efficiently — you only have 1 meter total

HEIGHT STRATEGY

- Tier system: build in 15–20cm sections, brace each one fully
- Taper upward — wider base, narrower top = lower center of gravity
- Heavier concentration of straws at the base = more stability
- Save 2–3 straws for emergency repairs or final reinforcement
- Unused straws count as tiebreaker — use nearly all of them

Types of Towers — Written Assessment Topics

Tower Types

The written assessment covers types and building of towers. Know these categories and examples — they will appear on your test.

BY FUNCTION

- Communication: cell towers, radio towers, TV broadcast towers
- Observation: lighthouses, fire towers, observation decks
- Religious: church steeples, minarets (Islamic), pagodas (Buddhist)
- Fortification: castle towers, watchtowers
- Water Towers: store water for distribution
- Transmission: support power lines

BY STRUCTURE

- Freestanding: supported only by their own weight (e.g., Eiffel Tower)
- Guyed: supported by cables anchored to the ground (e.g., radio masts)
- Lattice: framework of interconnected beams (e.g., transmission towers)
- Monopole: single vertical pole (e.g., some cell towers)

KEY CONCEPTS

- Stability: withstanding wind, earthquakes, and gravity
- Center of Gravity: point where weight is balanced — must stay over base
- Structural Integrity: ability to withstand loads without collapsing
- Materials Science: steel, concrete, wood — each has different strength/weight
- Foundation: must support total tower weight + live loads

🌐 Famous Towers — Know These for the Written Test

Famous

🗼 Eiffel Tower • Paris, France • 330m

Type: Freestanding lattice iron • 1889

Open lattice reduces wind resistance. Uses ~7,300 tons of iron — but if melted into a flat disk the same radius, it would be only 6cm thick. Icon of structural efficiency.

🏛️ Leaning Tower of Pisa • Pisa, Italy • 56m

Type: Freestanding masonry • 1372

Uneven soft soil caused one side to sink. Famous for ~4° lean. Stabilized with counterweights in 2001. Shows that foundations are as important as the structure.

📡 CN Tower • Toronto, Canada • 553m

Type: Hollow reinforced concrete • 1976

Tallest freestanding structure in Western Hemisphere. Tripod base for maximum stability. Hollow concrete tube — hollow is stiffer than solid for same weight.

🏙️ Burj Khalifa • Dubai, UAE • 828m

Type: Y-shaped steel/concrete • 2010

World's tallest building. Y-shaped floor plan distributes wind and torsion. Tapers as it rises — reduces wind force on upper floors.

⛪ Church Steeple • Various • Varies

Type: Religious fortification • Medieval+

Tall pointed spire reduces wind load. Acts as landmark. Lead or copper cladding protects from weather. Height symbolizes reaching toward the sky.

📶 Radio/Cell Tower • Worldwide • Varies

Type: Guyed lattice or monopole • Modern

Guyed towers use cables anchored to ground — allows taller, lighter structures. Lattice design minimizes wind resistance. Monopoles are self-supporting single poles.

🏆 Competition Strategy — 45 Minutes to Win

Strategy

WRITTEN TEST (simultaneous)

1. Done at the SAME TIME as building — divide roles
2. Partner 1: writes the test solo. Partner 2: leads the build
3. OR: both work on test first (10 min), then both build (35 min)
4. Review tower types, stability, and center of gravity before competition
5. Know the scoring formula: both height AND assessment count equally (50/50)

BUILD PHASE

1. Start with base: 4 straws in a square + 2 diagonals = first tier
2. Add 2nd tier before making it taller — bracing each level is key
3. Check vertical alignment every 10cm — a lean gets worse as you go up
4. Use tape sparingly but completely — run out = weaker upper tower
5. Reserve 2 straws for late-stage repairs or final reinforcement

MAXIMIZE YOUR SCORE

1. Both height AND assessment are worth 50 pts each — study both!
2. Use nearly all straws: unused straws are a tiebreaker disadvantage
3. Plumb (straight) towers beat leaning towers — measure against a wall
4. The 5×5cm block must fit your top platform — build it in from the start
5. After placing mass: step back and watch — do NOT touch the tower

📖 Key Vocabulary & Fast Facts

Vocab

Compression:	A pushing/squeezing force — straw columns are under compression from the 300g load above
Tension:	A pulling/stretching force — diagonal braces on one side go into tension when load shifts
Shear:	Sideways force sliding layers past each other — unbraced rectangles fail in shear
Buckling:	When a slender column under compression suddenly bends sideways and collapses
Triangulation:	Dividing a structure into triangles to make it rigid — the core straw tower principle
Center of Gravity:	Point where total weight appears to act — must stay above the base footprint

Stability:	Ability to remain upright; improved by wide base and low center of gravity
Structural Integrity:	Ability to carry loads without collapsing — what your tower is scored on
Lattice Tower:	Tower made of a framework of interconnected beams — e.g., Eiffel Tower
Freestanding Tower:	Supported entirely by its own weight and base — no cables or anchors needed
Guyed Tower:	Tower supported by cables anchored to the ground — allows taller, lighter designs
Load Path:	Route that forces travel from the top of the tower down to the ground